Strengths of Diffusion- and Flow- based Editing

Goal #2:
Instruction following
Editing capability

Goal #1:
Fidelity /
Reconstruction

Diffusion Models:
ODE Inversion (2021)

- **Rich diversity / fixability**

Source image Output image

User instruction:
Add ***a glass*** next to ***the cookie***

Rectified Flows:
Delta Velocity Injection (2024)

- **Fine-grained control**

Source image Output image

User instruction: (minimal modification)
Replace "***CAFE***" with "***FREE***"

Core Idea is Better Tradeoff

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Goal #1:
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DMs: ODE Inversion

- **Rich diversity/fixability**

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User instruction:
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RFs: Delta Velocity Injection

- **Fine-grained control**

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User instruction:
Replace "***CAFE***" with "***FREE***"